

Sentences of interpretation for confidence intervals

MATH 213

Difference of means - observational study using random sampling

I am 95% confident that the mean age of first marriage among 2006 US residents who have ever been married is between 0.76 and 1.64 years older than the mean age of first marriage among 1972 US residents who have ever been married.

Difference of proportions - experiment using random assignment

We can be 95% confident that the proportion of bank managers that would promote if all were given a male file to judge is between 0.0495 and 0.5339 higher than if they were all given female files to judge.

Difference of means - experiment using random assignment

We are 95% confident that if all participants were placed in the incentive group they would lose between 5.57 and 17.9 pounds more on average than if all were placed in the control group.